Smart Renewable Energy Management System with Adaptive Load Shedding

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***Abstract*—This paper presents a MATLAB/Simulink study of a solar photovoltaic microgrid for a residential block of 12 flats. The design combines a 38.4-kW photovoltaic array built from two 19.2-kW units, perturb-and-observe maximum power point tracking, a 400-V DC bus, a battery and supercapacitor hybrid energy storage system, low-pass-filter power sharing with state-of-charge-aware limits, a four-state Stateflow load manager with three priority tiers, and a five-level cascaded H-bridge inverter. An appliance-level load model gives a daily demand of 211.6 kWh, of which 117.8 kWh is firm, with a 16.5-kW evening peak. For a clear-day Coimbatore weather profile, the detailed switching model of one photovoltaic unit produces about 15.4 kW at peak and 101.6 kWh per day, so two units supply about 203 kWh per day, or 96 percent of the modelled demand. The DC-bus voltage stays within 1.4 percent of its reference, and the five-level inverter produces output voltage and current total harmonic distortion of 0.93 percent and 0.26 percent up to the 20th harmonic. Storage, load-manager and integrated-model results are reported as preliminary subsystem tests, and the parameter differences that must be resolved before a full-day closed-loop run are identified.**

***Keywords*—Solar photovoltaic system, maximum power point tracking, DC microgrid, hybrid energy storage, supercapacitor, energy management system, adaptive load shedding, cascaded H-bridge inverter.**

I. Introduction

The transition towards low-carbon electricity has moved generation closer to the consumer, and residential microgrids built around rooftop solar photovoltaic (PV) arrays are now a practical way to supply apartment blocks. A microgrid groups local sources, storage and loads under a common controller so that the cluster can be operated as a single controllable unit [1], [2]. Its control is normally organised in layers, from fast converter-level loops to slower supervisory energy management [3], [4]. For a PV-dominated residential microgrid, the supervisory layer has to deal with one central difficulty: solar output follows irradiance and cell temperature, peaking near midday, whereas household demand peaks in the morning and, more strongly, in the evening.

At the source, the PV array must be operated at its maximum power point (MPP) while irradiance and temperature change. Accurate single-diode array models allow this behaviour to be reproduced in simulation [5], and maximum power point tracking (MPPT) algorithms adjust the operating voltage of a DC–DC converter to follow the MPP. Of the many MPPT methods available [6], perturb and observe (P&O) remains the most widely used because it needs only voltage and current measurements; its step size and sampling interval decide the trade-off between tracking speed and steady-state oscillation [7].

Storage is needed to move the midday surplus to the evening deficit. Battery energy storage systems (BESS) have high energy density but limited power density, and frequent high-rate current changes caused by cloud transients and load steps accelerate their ageing. Hybrid energy storage systems (HESS) that pair a battery with a supercapacitor address this by giving the slow part of the power mismatch to the battery and the fast part to the supercapacitor [8], [9]. Power sharing in a HESS has been obtained by optimisation [10], model predictive control [11], low-pass filtering with fixed or variable time constants [12], and by routing the high-frequency component of the DC-bus voltage controller, together with the uncompensated battery current, to the supercapacitor converter [13]. State-of-charge (SoC) and operating-mode logic is then added so that neither device is pushed beyond its limits [14], [15].

When generation and stored energy together cannot meet demand, the remaining option is to reduce the load in a controlled way. Residential loads differ widely in importance: refrigeration, communication and emergency lighting are critical, lighting and fans are essential, and air conditioners, water heaters and laundry are deferrable. A priority-based load manager can shed the lowest tier first and restore it when storage recovers, which keeps critical services supplied for longer than an undifferentiated cut-off. Load shedding of this kind is also an established part of islanded microgrid control [16]. Supervisory schemes of this kind have been combined with MPPT and storage control in hybrid and residential microgrids [17], [18].

Finally, the DC bus has to supply standard AC appliances.

Multilevel inverters synthesise a stepped output from several DC sources and give lower harmonic distortion and lower device stress than two-level inverters [19], [20]. The cascaded H-bridge (CHB) topology is modular [21], and with phase-shifted carrier modulation [22] a five-level output can be obtained from two series-connected cells.

This work models these stages together for a block of 12 flats in Coimbatore, India. The main contributions are:

- an appliance-level, three-tier load model for 12 flats and a PV array sized against its firm daily energy, connected through a boost converter with P&O MPPT to a common $400\,\mathrm{V}$ DC bus under a 24-hour weather profile;
- a battery–supercapacitor HESS with a low-pass-filter energy management system (EMS), SoC-aware power limits, and a combined current controller that passes fast DC-bus transients to the supercapacitor;
- a four-state Stateflow load manager with hysteresis that sheds and restores load tiers according to storage state and power deficit; and
- a five-level CHB inverter evaluated through waveform and harmonic analysis, together with a time-compressed integrated model of the complete microgrid.

II. Literature Summary

Hilmi *et al.* presented a finite-state-machine EMS for a standalone PV system with battery–supercapacitor storage, in which the operating mode is selected from the SoC of both devices [14]. Dwivedi and Ray studied energy management of a grid-connected microgrid with HESS, including power exchange and sharing between the battery and supercapacitor [15]. Khemissi *et al.* combined MPPT with a supervisory strategy in a wind/PV/battery smart microgrid [17], and Rajkumar and Gangatharan proposed an AC/DC residential microgrid with PV and storage that reduces unnecessary conversion stages [18]. On power sharing, Choi *et al.* formulated an optimisation that reduces battery-current fluctuation and supercapacitor loss [10], Wu *et al.* used low-pass filtering with a variable time constant [12], Hredzak *et al.* applied model predictive control [11], and Kollimalla *et al.* assigned the high-frequency part of the DC-link controller output and the uncompensated battery current to the supercapacitor [13]. Device models for the battery [23] and the double-layer capacitor [24] used in common simulation libraries are also well established.

Together, these studies cover PV integration, hybrid storage and supervisory control. The present project applies them to a specific residential design problem: a switching-level PV model driven by a weather profile, a tiered appliance-level demand model, SoC-aware HESS power allocation, and a priority load manager, all in one Simulink framework.

III. Research Gap

PV output is intermittent and does not coincide with residential demand. A battery-only interface is exposed to fast power variations, and a fixed load-shedding rule does not respond to changes in generation, demand or storage state. The literature reviewed above treats HESS power sharing, DC-bus regulation, MPPT and residential energy management mostly as separate problems. The gap addressed here is to bring these functions together in one simulation of a realistic residential block: filtering the power mismatch into slow and fast components, constraining battery operation by SoC, and coordinating any remaining deficit with a priority-based load manager.

The project objectives are to:

- model the residential demand of 12 flats by appliance and priority tier;
- model a PV source and its boost-converter interface, and track the MPP with P&O under a time-varying weather profile;
- connect PV generation to a regulated $400\,\mathrm{V}$ DC bus;
- integrate battery–supercapacitor storage with SoC-aware power sharing and priority load shedding; and
- evaluate generation, energy balance, DC-bus voltage, storage response, inverter output quality and load-tier management.

IV. Block Diagram Explanation

Fig. 1 shows the system architecture. On the generation path, irradiance and ambient-temperature profiles drive the PV array model, and the boost converter with P&O MPPT transfers the available PV power to a common $400\,\mathrm{V}$ DC bus. The design uses two identical $19.2\,\mathrm{kW}$ PV units ($38.4\,\mathrm{kW}$ in total). The detailed switching model represents one unit, and the two-unit daily curve is obtained by doubling the single-unit power trace rather than by simulating two electrical branches.

The battery and supercapacitor are connected to the DC bus in parallel through bidirectional DC–DC converters. The battery takes the slow, sustained part of the power mismatch and the supercapacitor the fast part. The DC bus feeds a five-level CHB inverter and an LC filter, which supply the AC loads of the 12 flats. A supervisory path uses PV power, load demand, DC-bus voltage and storage state: the EMS generates the storage references, and the Stateflow load manager sheds or restores load tiers when storage or power limits are reached.

V. Methodology

The methodology consists of building the demand model, sizing and modelling the PV source, sizing and controlling the HESS, designing the supervisory EMS and load manager, modelling the CHB inverter, and evaluating each subsystem and the integrated model in MATLAB/Simulink. All parameters are set by the initialisation script `init_microgrid.m`.

A. Residential Load Model

The block contains $N = 12$ identical flats and a set of common loads. Each appliance a is described by its tier, quantity n_a, rated power P_a and a schedule of duty factors $\delta_a(h)$ for each hour h. The hourly demand of tier k is

$$P_k(h) = \sum_{a \in \mathcal{T}_k} n_a P_a \delta_a(h), \qquad k = 1, 2, 3, \tag{1}$$

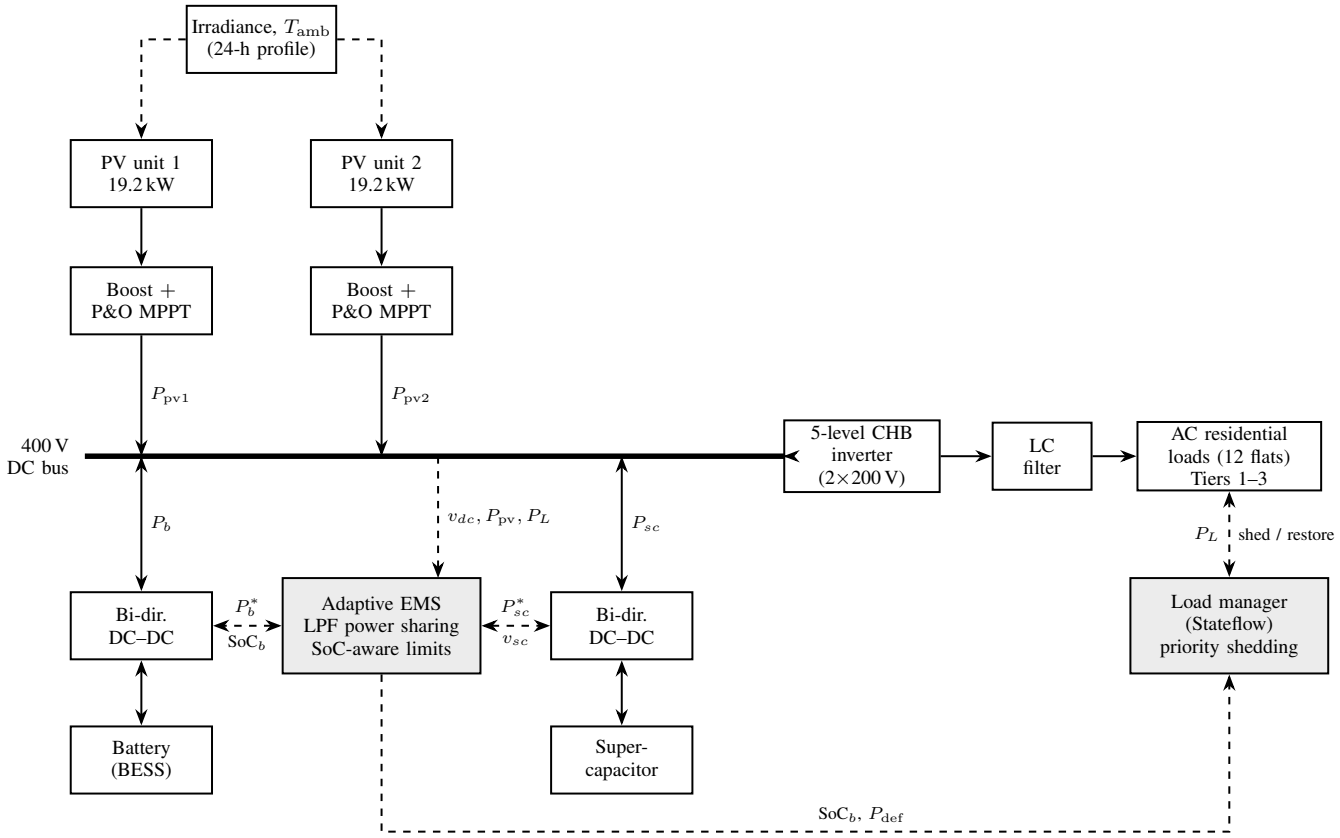


Fig. 1. System architecture. The two PV branches show the design configuration; the detailed PV switching simulation is for one unit and the two-unit generation curve is obtained by scaling its trace. Solid lines denote power flow; dashed lines denote measurement and control signals.

TABLE I
RESIDENTIAL LOAD PRIORITY TIERS (12 FLATS)

Tier	Representative loads	Peak (kW)	Energy (kWh/d)
1 (critical)	Refrigerators, Wi-Fi routers, CCTV/security, common and emergency lighting	1.45	28.8
2 (essential)	Flat lighting, ceiling fans, TV and plug loads, kitchen appliances, lift	9.87	89.0
3 (deferrable)	Split air conditioners, geysers, washing machines, irons, water pump	9.00	93.8
Firm	Tiers 1–2	–	117.8
Total	Tiers 1–3	16.5[a]	211.6

[a]Coincident peak of the total profile (21:00–23:00).

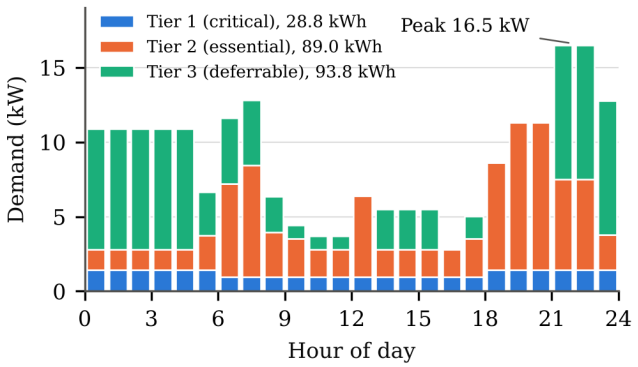


Fig. 2. Hourly demand of the 12-flat block by priority tier, computed from the appliance schedules in the initialisation script. Daily energy per tier is given in the legend.

and the daily tier energy is $E_k = \sum_{h=0}^{23} P_k(h) \times 1\,\mathrm{h}$. The schedule is calibrated so that each flat uses about $15.5\,\mathrm{kWh/day}$ on a summer design day, of which air conditioning is the largest share. Table I lists the tiers and Fig. 2 shows the resulting profile. The total demand is $211.6\,\mathrm{kWh/day}$, the firm (Tier 1 and Tier 2) demand is $117.8\,\mathrm{kWh/day}$, and the peak is $16.5\,\mathrm{kW}$ between 21:00 and 23:00, when PV output is zero. The firm demand between 18:00 and 06:00 is $67.8\,\mathrm{kWh}$, which sets the overnight storage requirement.

B. PV Array Sizing

The PV array is sized to supply the firm daily energy. Using a peak-sun-hour value of $\mathrm{PSH} = 4.8\,\mathrm{h/day}$, a temperature derating factor $\eta_{\mathrm{temp}} = 0.88$, a soiling, mismatch and wiring factor $\eta_{\mathrm{soil}} = 0.95$ and a system factor $\eta_{\mathrm{sys}} = 0.85$, the

TABLE II
PV ARRAY AND BOOST-CONVERTER PARAMETERS

Parameter	Value
Module rating / V_{mp} / I_{mp}	400 W / 40 V / 10 A
Module V_{oc} / I_{sc}	48 V / 10.5 A
Series modules × parallel strings	6 × 8 per unit
PV rating per unit / total (two units)	19.2 kW / 38.4 kW
Nominal string V_{mp} / V_{oc}	240 V / 288 V
DC-bus reference	400 V
Boost inductance	0.30 mH
Input / output capacitance (Simulink block)	100 µF / 100 µF
Switching frequency	20 kHz
MPPT method / duty step ΔD	P&O / 0.002
Duty-cycle window (MPPT mode)	0.36–0.60
Night parking	$D = 0.42$ when $P_{\mathrm{pv}} < 50$ W

minimum rating is

$$P_{\mathrm{PV,min}} = \frac{E_{\mathrm{firm}}}{\mathrm{PSH}\,\eta_{\mathrm{temp}}\,\eta_{\mathrm{soil}}\,\eta_{\mathrm{sys}}} = \frac{118}{4.8 \times 0.88 \times 0.95 \times 0.85} \approx 34.6\,\mathrm{kW}. \tag{2}$$

An installed rating of 38.4 kW is selected, which gives an 11% margin over this estimate.

C. PV Module and Array Configuration

Each module is rated 400 W with $V_{\mathrm{mp}} = 40$ V, $I_{\mathrm{mp}} = 10$ A, $V_{\mathrm{oc}} = 48$ V, $I_{\mathrm{sc}} = 10.5$ A and 72 cells. Six modules are connected in series per string and eight strings in parallel per unit, so

$$P_{\mathrm{unit}} = 6 \times 8 \times 400\,\mathrm{W} = 19.2\,\mathrm{kW}, \tag{3}$$

$$V_{\mathrm{mp,string}} = 6 \times 40\,\mathrm{V} = 240\,\mathrm{V}, \tag{4}$$

$$V_{\mathrm{oc,string}} = 6 \times 48\,\mathrm{V} = 288\,\mathrm{V}, \tag{5}$$

with a nominal unit current of $8 \times 10\,\mathrm{A} = 80\,\mathrm{A}$. The array is represented by the single-diode PV Array block, whose fitted parameters include a series resistance of $0.200\,\Omega$ and a shunt resistance of $821.8\,\Omega$ per module, and temperature coefficients of $-0.361\%/^{\circ}\mathrm{C}$ for V_{oc} and $+0.102\%/^{\circ}\mathrm{C}$ for I_{sc} [5]. These ratings refer to standard test conditions (STC); under the weather profile the output is lower because irradiance is below $1000\,\mathrm{W/m^2}$ and the cells are hotter than $25\,^{\circ}\mathrm{C}$. Table II summarises the PV-side parameters.

D. Weather and Cell-Temperature Model

A 24-hour clear-day profile representative of Coimbatore is applied to the PV model. Irradiance rises after sunrise, peaks at about $950\,\mathrm{W/m^2}$ near midday and falls to zero after sunset. The profile is read from look-up tables and held for 15-minute intervals by a zero-order hold. The cell temperature is estimated from the ambient temperature and irradiance G with the nominal operating cell temperature (NOCT) relation

$$T_{\mathrm{cell}} = T_{\mathrm{amb}} + \frac{\mathrm{NOCT} - 20}{800}\,G, \qquad \mathrm{NOCT} = 45\,^{\circ}\mathrm{C}, \tag{6}$$

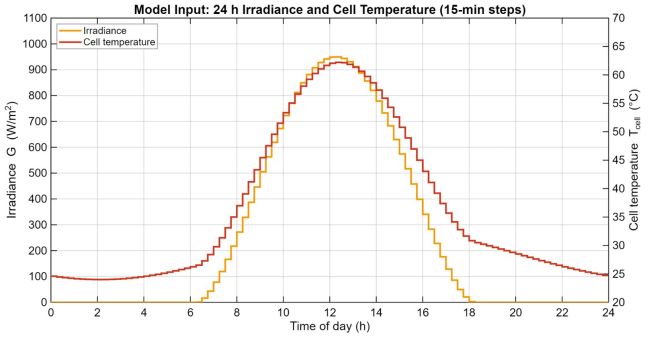


Fig. 3. Twenty-four-hour irradiance and modelled PV cell-temperature inputs used in the PV simulation.

with temperatures in °C and G in $\mathrm{W/m^2}$. Fig. 3 shows both inputs. For the supplementary daily-energy calculation, the MPP power is approximated as

$$P_{\mathrm{MPP}} \approx P_{\mathrm{STC}}\,\frac{G}{1000}\,[1 + \gamma\,(T_{\mathrm{cell}} - 25)]\,, \tag{7}$$

where $\gamma = -0.004/^{\circ}\mathrm{C}$ is the power temperature coefficient used in the profile script.

E. Boost Converter and P&O MPPT

The string MPP voltage is about 240 V at STC while the DC bus is held at 400 V, so a boost converter is used [25]. In continuous conduction the PV voltage and duty ratio are related by

$$V_{\mathrm{pv}} = (1 - D)\,V_{\mathrm{dc}}, \qquad D_{\mathrm{STC}} = 1 - \frac{240}{400} = 0.40. \tag{8}$$

Lowering D therefore raises the PV voltage. The P&O algorithm [7] samples V_{pv} and I_{pv} every 0.1 ms, computes $\Delta P_k = P_k - P_{k-1}$ and $\Delta V_k = V_k - V_{k-1}$, and updates the duty ratio as

$$D_{k+1} = \begin{cases} D_{\mathrm{park}} = 0.42, & P_k < 50\,\mathrm{W}, \\ D_k, & \Delta P_k = 0, \\ D_k - \Delta D, & \Delta P_k\,\Delta V_k > 0, \\ D_k + \Delta D, & \text{otherwise}, \end{cases} \tag{9}$$

with $\Delta D = 0.002$, and then limits D_{k+1} to 0.36–0.60. This window corresponds to PV voltage commands of 160–256 V. It covers every MPP of the day (about 195–245 V) but stays below the open-circuit voltage, so the tracker cannot drift to open circuit when irradiance rises at sunrise. The night parking state holds the operating point near $0.8\,V_{\mathrm{oc}}$ when the PV power is negligible.

The PV block also includes a curtailment mode. When the battery SoC reaches 98%, a switch replaces the MPPT duty with the output of a PI controller ($K_p = 0.001$, $K_i = 0.05$) that regulates the DC bus at 400 V, with the duty limited to 0.05–0.85. The PV unit then supplies only the power that the bus can absorb. A 20 kHz sawtooth carrier compared with the duty command generates the gate signal.

F. Hybrid Energy Storage Sizing and Modelling

1) Battery: The battery supplies the firm overnight demand plus a reserve. With $E_{\text{night}} \approx 68\,\text{kWh}$, a reserve $E_{\text{res}} = 30\,\text{kWh}$, a usable depth of discharge of 0.8 and a discharge efficiency $\eta_d = 0.95$,

$$E_{\text{bat}} = \frac{E_{\text{night}} + E_{\text{res}}}{\text{DoD}\,\eta_d} = \frac{68 + 30}{0.8 \times 0.95} \approx 129\,\text{kWh}, \qquad (10)$$

and a 130 kWh, 240 V lithium-ion bank (541.7 Ah) is chosen. Within the 20–90% SoC operating window this leaves 91 kWh of usable energy, which exceeds the firm night demand. The battery is represented by the generic lithium-ion block based on the model of Tremblay and Dessaint [23], with an initial SoC of 80%. Its SoC follows coulomb counting:

$$\text{SoC}_b(t) = \text{SoC}_b(0) - \frac{100}{3600\,Q_b}\int_0^t i_b(\tau)\,d\tau, \qquad (11)$$

where Q_b is the capacity in ampere-hours. Because a whole day is compressed into a few simulated seconds (Section V-K), the capacity entered in the block must be scaled by the same factor,

$$Q_{\text{sim}} = Q_{\text{real}}\,\frac{t_{\text{hour}}}{3600}, \qquad (12)$$

so that the SoC swing over one compressed day equals that over one real day.

2) Supercapacitor: The supercapacitor is designed as a 10 F, 250 V bank. Its stored energy is

$$E_{sc} = \tfrac{1}{2}C_{sc}v_{sc}^2 = \tfrac{1}{2} \times 10 \times 250^2 = 312.5\,\text{kJ}. \qquad (13)$$

Discharging to half of the rated voltage releases 75% of this energy, about 234 kJ, enough to supply a 20 kW transient for roughly 12 s. It is represented by the Stern-type supercapacitor block [24].

3) DC link and converters: The DC-link capacitance is sized to supply a 20 kW step for 10 ms with a 5% voltage dip:

$$C_{dc} = \frac{2P\,\Delta t}{V_{\text{dc}}^2 - (0.95V_{\text{dc}})^2} = \frac{2 \times 20\,000 \times 0.01}{400^2 - 380^2} \approx 25.6\,\text{mF}. \qquad (14)$$

Each storage element connects to the bus through a non-isolated bidirectional half-bridge converter with two IGBT/diode switches: 0.30 mH for the battery and 0.355 mH for the supercapacitor. Table III lists the design values in the initialisation script next to the values entered in the supplied Simulink blocks; the differences are discussed in Section VI-G.

G. Adaptive Energy Management System

The EMS works on the power mismatch between demand and PV generation,

$$P_{\text{mis}}(t) = P_L(t) - P_{\text{pv}}(t), \qquad (15)$$

which is positive in deficit and negative in surplus. A first-order discrete low-pass filter extracts its slow component for the battery:

$$P_{b,\text{base}}[k] = \alpha\,P_{b,\text{base}}[k-1] + (1-\alpha)\,P_{\text{mis}}[k], \quad \alpha = e^{-T_s/\tau}, \qquad (16)$$

TABLE III
STORAGE AND DC-LINK PARAMETERS: DESIGN SCRIPT AND SIMULINK BLOCK

Parameter	Design (script)	Simulink block
Battery type / nominal voltage	Li-ion / 240 V	Li-ion / 240 V
Battery capacity	130 kWh (541.7 Ah)	$50\,t_{\text{hour}}/3600$ Ah
Battery power limit	25 kW	–
Initial SoC / window	80% / 20–90%	80% / –
Charge / discharge eff.	0.95 / 0.95	–
SC capacitance	10 F	20/3600 F
SC rated / initial voltage	250 V / 250 V	250 V / 240 V
DC-link capacitance	25 mF	300 µF (init. 400 V)
Storage converter PWM	20 kHz	16 kHz
EMS LPF time constant	2 ms	0.2 s

TABLE IV
EMS OPERATING MODES

Mode	Condition	Action
M1	PV surplus, $\text{SoC}_b < 98\%$	Charge storage, charging derated above 90% SoC
M2	PV surplus, $\text{SoC}_b \geq 98\%$	PV switches to DC-bus voltage control (curtailment)
M3	Deficit, storage available	Battery supplies slow component; SC supplies residual
M4	Deficit, $\text{SoC}_b < 30\%$	Battery discharge derated to 20%; load manager sheds tiers
M5	PV negligible (night)	MPPT parked; storage supplies loads within limits

where T_s is the sample time and τ the filter time constant. An adaptive function then derates the battery request near its SoC limits:

$$P_b^* = \begin{cases} 0.2\,P_{b,\text{base}}, & \text{SoC}_b < 30\% \text{ and } P_{b,\text{base}} > 0, \\ 0.2\,P_{b,\text{base}}, & \text{SoC}_b > 90\% \text{ and } P_{b,\text{base}} < 0, \\ P_{b,\text{base}}, & \text{otherwise.} \end{cases} \qquad (17)$$

The supercapacitor receives the remainder, $P_{sc}^* = P_{\text{mis}} - P_b^*$, which contains the fast part of the mismatch and any battery power withheld by (17). The power references are converted to current references,

$$i_b^* = \frac{P_b^*}{v_b}, \qquad i_{sc,\text{EMS}}^* = \frac{P_{sc}^*}{v_{sc}} - i_{sc}, \qquad (18)$$

with the measured voltages limited to 1–1000 V to avoid division by zero. The unfiltered mismatch is also passed to the load manager as the power deficit P_{def}. Table IV summarises the resulting operating modes.

H. Combined Battery/Supercapacitor Current Control

The EMS references are tracked by a combined controller that also regulates the DC bus, following the structure of [13]. A PI controller on the bus-voltage error $e_v = V_{\text{dc}}^* - v_{\text{dc}}$ gives the total current demand i_{tot}^*. A first-order low-pass filter with a 15 ms time constant removes its slow part, and the high-

TABLE V
CONTROLLER GAINS IN THE SUPPLIED SIMULINK MODEL

Loop	K_p	K_i
DC-bus voltage (outer)	1.477	3077
Battery current (inner)	0.043	0.65
Supercapacitor current (inner)	0.45	14 800
PV constant-voltage (curtailment)	0.001	0.05
HF split filter: first-order low-pass, 15 ms		

frequency remainder is added to the uncompensated battery current error and referred to the supercapacitor side:

$$i_{sc}^{*} = \left[(i_{\mathrm{tot}}^{*} - \bar{i}_{\mathrm{tot}}^{*}) + (i_b^{*} - i_b)\right]\frac{v_b}{v_{sc}} + i_{sc,\mathrm{EMS}}^{*}, \tag{19}$$

where $\bar{i}_{\mathrm{tot}}^{*}$ is the filtered demand. The battery current loop tracks i_b^{*} only, so fast bus disturbances are carried by the supercapacitor and any current the battery cannot follow is passed to the supercapacitor as well. Each current PI output is limited to 0–1 and drives a 16 kHz DC–DC PWM generator that switches the two converter legs complementarily. The tuned gains are listed in Table V. The initialisation script gives analytical starting values from $K_{p,i} = 2\pi f_{ci} L / V_{\mathrm{dc}}$ and $K_{i,i} = K_{p,i}\, 2\pi f_{ci}/10$ with $f_{ci} = f_{sw}/10$; the final block values were tuned with the PID Tuner.

I. Cascaded H-Bridge Inverter

A single-phase five-level CHB inverter [19], [20] is built from two H-bridge cells (eight IGBT/diode switches in total) supplied from a split DC link of two 2200 µF capacitors charged to 200 V. With the cell outputs in series, the output takes the levels -400, -200, 0, $+200$ and $+400$ V:

$$v_o = \sum_{j=1}^{2} S_j\, V_{\mathrm{dc},j}, \qquad S_j \in \{-1, 0, +1\}. \tag{20}$$

Gate signals are produced by phase-shifted carrier PWM [22]. A 50 Hz sinusoidal reference ($\omega = 314\,\mathrm{rad/s}$) is compared with four 1 kHz triangular carriers displaced by 90°, so the output voltage ripple appears at four times the carrier frequency. An LC filter removes the remaining switching components. On the AC side, each load tier is modelled as a controlled current source,

$$i_k(t) = \frac{P_k\, v(t)}{V_{\mathrm{rms}}^2}, \qquad V_{\mathrm{rms}} = 230\,\mathrm{V}, \tag{21}$$

which behaves as a resistive load that draws P_k at rated voltage, and is connected through an ideal switch controlled by the load manager.

J. Priority-Based Adaptive Load Shedding

The load manager is a Stateflow chart with four states. Its output L selects which tiers are connected (Table VI): tier k is connected when $L \le 3-k$, that is $u_1 = (L \le 2)$, $u_2 = (L \le 1)$ and $u_3 = (L = 0)$. The transitions use the battery SoC (as a fraction), its rate of change, and the power deficit P_{def} from the EMS. The thresholds are $\theta_1 = 0.40$, $\theta_2 = 0.30$, $\mathrm{SoC}_{\min} = 0.20$ and a restoration hysteresis $d = 0.08$. Table VII gives the guards, and Fig. 4 shows the chart.

TABLE VI
LOAD-MANAGER STATES

State	Tier 1	Tier 2	Tier 3
$L = 0$ Normal	On	On	On
$L = 1$ Shed deferrable	On	On	Off
$L = 2$ Critical only	On	Off	Off
$L = 3$ Outage	Off	Off	Off

TABLE VII
LOAD-MANAGER TRANSITION GUARDS

Transition	Guard
$0 \to 1$	$\mathrm{SoC} < \theta_1$ or ($\mathrm{SoC} < \theta_1 + 0.05$ and $\dot{\mathrm{SoC}} < -0.15$)
$1 \to 2$	$\mathrm{SoC} < \theta_2$
$2 \to 3$	$\mathrm{SoC} \le \mathrm{SoC}_{\min}$ or $P_{\mathrm{def}} > 22$
$3 \to 2$	$\mathrm{SoC} > \mathrm{SoC}_{\min} + 0.10$ and $P_{\mathrm{def}} \le 10$
$2 \to 1$	$\mathrm{SoC} > \theta_2 + d$
$1 \to 0$	$\mathrm{SoC} > \theta_1 + d$

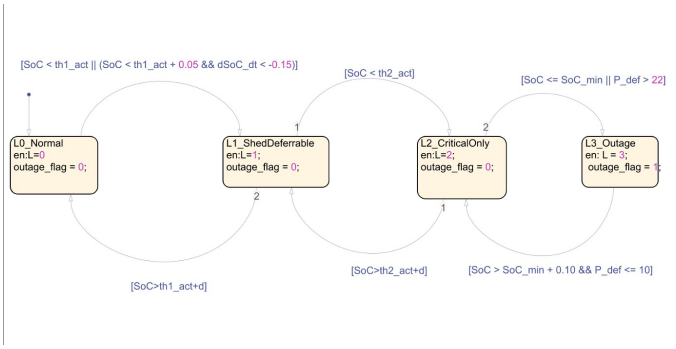


Fig. 4. Stateflow load-management chart with normal, deferrable-shedding, critical-only and outage states.

Shedding is anticipatory: the deferrable tier is dropped not only when SoC falls below θ_1, but also when it is within 0.05 of θ_1 and falling faster than 0.15 per unit time. Every restoration requires the SoC to recover by d above the shedding threshold, which prevents chattering around a threshold. The load function also tracks the unserved deferrable energy,

$$E_{\mathrm{def}} \leftarrow \min\Big(E_{\max},\; E_{\mathrm{def}} + P_3(1-u_3)\,\Delta h - u_3\,\frac{E_{\mathrm{def}}}{T_c}\,\Delta h\Big), \tag{22}$$

where Δh is the real time per solver step, $T_c = 2\,\mathrm{h}$ is the recovery time constant and $E_{\max} = 10\,\mathrm{kWh}$. This records how much deferred demand must be served after the tier is restored. Random variability of 3%, 10% and 15% can be added to Tiers 1, 2 and 3 respectively.

K. Simulation Procedure and Performance Evaluation

All power stages are simulated in discrete mode with a fixed step of 1 µs. A full switching simulation of 24 real hours is not practical, so time is compressed: t_{hour} simulated seconds represent one real hour. The PV unit model uses $t_{\mathrm{hour}} = 0.05\,\mathrm{s}$

(1.2 s per day). The integrated model uses $t_{\text{hour}} = 0.12\,\text{s}$ (2.88 s per day), runs the EMS and Stateflow chart every 2 ms (one real minute), and updates loads and weather every 30 ms (15 real minutes). Table VIII lists the test conditions.

The following indices are used. The daily PV energy is $E_{\text{pv}} = \int P_{\text{pv}}\,dt$ expressed in real hours, and the energy coverage ratio is $\rho = E_{\text{pv}}/E_{\text{load}}$. The DC-bus deviation is

$$\Delta V_{\text{dc}}(\%) = \frac{\max|v_{\text{dc}} - V_{\text{dc}}^*|}{V_{\text{dc}}^*} \times 100. \tag{23}$$

Inverter output quality is measured by the total harmonic distortion,

$$\text{THD} = \frac{\sqrt{\sum_{h=2}^{\infty} X_h^2}}{X_1} \times 100\%, \tag{24}$$

where X_h is the RMS value of the h-th harmonic of voltage or current, and the result is compared with IEEE Std 519 [26].

VI. Results and Discussion

This section reports the four tests in Table VIII. Table IX at the end of the section collects the quantitative indices.

A. Test 1: Weather Input and PV Generation

Fig. 3 shows the applied weather input. Irradiance is zero until about 06:30, peaks at about $950\,\text{W/m}^2$ shortly after 12:00, and returns to zero at about 18:00. The cell temperature computed from (6) follows the irradiance and reaches about 62 °C at noon, while at night it falls back to the ambient level of 24–30 °C.

Fig. 5 shows the response of one PV unit over the day. Power is produced between about 06:45 and 18:00 and peaks at about 15.4 kW near 12:00, which is 80% of the 19.2 kW STC rating. This shortfall agrees with (7): at $G = 950\,\text{W/m}^2$ and $T_{\text{cell}} = 62\,°\text{C}$,

$$P_{\text{MPP}} \approx 19.2 \times 0.95 \times [1 - 0.004\,(62 - 25)] \approx 15.5\,\text{kW}, \tag{25}$$

within 1% of the simulated peak. The PV current rises to about 78 A at noon, close to the nominal 80 A, while the PV voltage stays between about 200 and 225 V throughout the day, falling slightly at midday as the cells heat up.

The duty-ratio trace confirms each part of the P&O logic in (9). At night the duty ratio is parked at 0.42. At dawn (about 06:50–07:10) and dusk (about 17:35–18:00) it reaches the lower limit of 0.36, where the tracker raises the PV voltage towards the top of its window to find the MPP at low irradiance. During the day it rises from about 0.44 in the morning to 0.50 at noon. With $V_{\text{pv}} \approx 200$ V at noon, (8) gives $1 - 200/400 = 0.50$, which matches the simulated value. The daily energy is 101.6 kWh per unit.

The DC-bus voltage is 400 V at night and rises with PV output to a maximum of 405.5 V at noon, a deviation of 1.4%. Because the deviation follows the PV power profile, it reflects the finite stiffness of the bus in the single-unit test rather than a tracking error. The thicker bands in the voltage, current and duty traces are the P&O perturbation of $\pm\Delta D$ around the MPP.

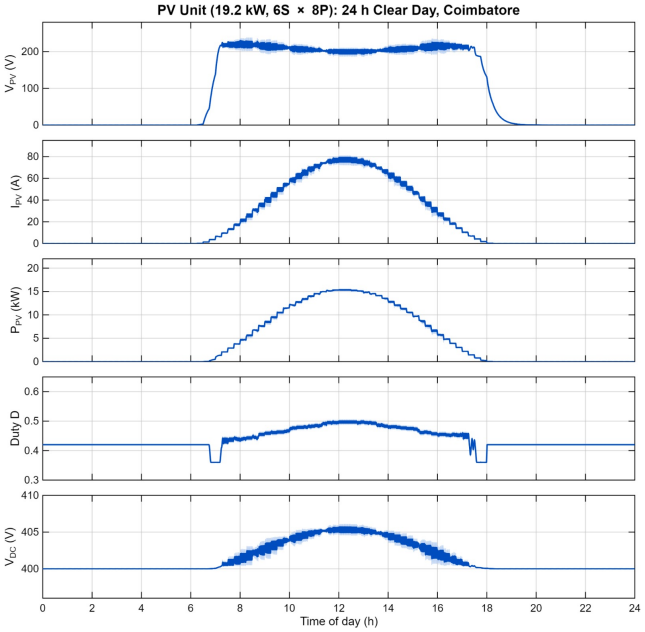


Fig. 5. Test 1: single-unit PV voltage, current, power, boost-converter duty ratio and DC-bus voltage over a 24-hour clear day.

B. Daily Energy Balance

Fig. 6 compares the two-unit PV profile with the demand. The array peaks at about 30.8 kW and delivers about 203 kWh/day, compared with 211.6 kWh/day of demand. The coverage ratio is therefore $\rho \approx 0.96$, and the array exceeds the 117.8 kWh/day firm demand by 72%. The daily shortfall of about 9 kWh is 4% of the demand and can be covered from storage or by deferring Tier 3 loads.

The timing of the two curves matters more than the totals. PV generation exceeds the total demand only between about 07:50 and 16:50, and the midday surplus reaches about 25 kW. Demand reaches its 16.5 kW peak at 21:00–23:00, after sunset, and the whole night load from 18:00 to 07:00 must come from storage. The midday surplus must therefore be stored to supply the 67.8 kWh of firm night demand, which the 91 kWh usable battery capacity from (10) can cover. Around noon the surplus also exceeds the 25 kW battery power limit, so the excess must be taken by the supercapacitor or, once the battery is full, curtailed by the constant-voltage mode of the PV converter.

C. Test 3: Load-Manager Response

Fig. 7 shows the load manager driven by the benchmark SoC profile, which falls from 0.80 to about 0.25, rises to about 0.94 and falls again to about 0.35. The top panel is the state L, and the two lower panels are the requested and active tier powers. The response follows the guards in Table VII:

- at about 0.73 s, SoC falls below $\theta_1 = 0.40$ and the chart moves to $L = 1$; the active Tier 3 power drops to zero while its requested power continues;
- at about 0.90 s, SoC falls below $\theta_2 = 0.30$ and the chart moves to $L = 2$, disconnecting Tier 2;

TABLE VIII
TEST CONDITIONS USED FOR THE SIMULATION RESULTS

Parameter	Test 1 – PV unit	Test 2 – CHB inverter	Test 3 – Load manager	Test 4 – Integrated
Model	One 19.2 kW unit, boost, P&O	Two-cell CHB with LC filter	Stateflow chart and load function	Full microgrid model
Time step	1 µs	1 µs	2 ms (EMS rate)	1 µs power, 2 ms EMS
Time mapping	0.05 s per real hour	Real time	0.12 s per real hour	0.12 s per real hour
Duration	1.2 s (24 h)	0.04 s shown (two cycles)	0–2.4 s	0–0.22 s (battery), 0.21–0.31 s (power)
Input	Clear-day G, T_{amb} (15 min)	200 V per cell, about 4.3 kW load	Benchmark SoC and $\dot{\mathrm{SoC}}$ profiles	Clear-day weather, workspace load profile
Signals recorded	V_{pv}, I_{pv}, P_{pv}, D, v_{dc}	v_o, i_o, cell V and I; FFT to 1 kHz	L, SoC, requested and active tier power	v_b, i_b, SoC; P_{pv}, P_L, P_{HESS}, P_{def}

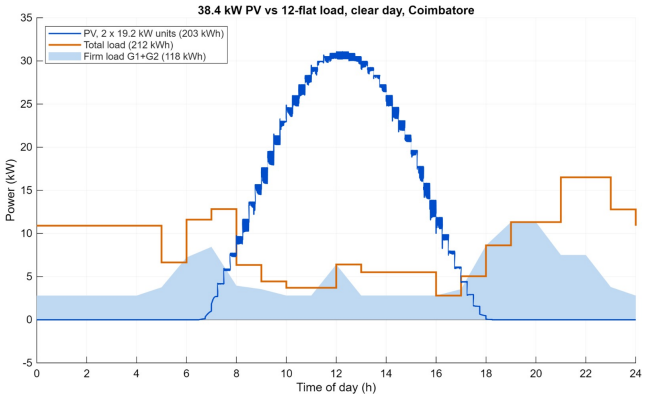


Fig. 6. Two-unit PV generation profile compared with the total and firm (Tier 1 + Tier 2) demand of the 12-flat block on a clear day.

- while SoC recovers, the chart returns to $L = 1$ at about 1.18 s (SoC $\approx 0.46 > \theta_2 + d$) and to $L = 0$ at about 1.20 s (SoC $\approx 0.50 > \theta_1 + d$), and all tiers are reconnected; and
- at about 2.22 s, SoC falls below 0.40 again and Tier 3 is shed a second time.

Tier 3 is always shed first and restored last, which confirms the intended priority order.

The trace also shows two unintended outage intervals ($L = 3$) at about 0.93–1.07 s and 1.13–1.15 s, during which Tier 1 is also disconnected. The SoC is about 0.28–0.31 at these instants, above $\mathrm{SoC}_{\min} = 0.20$, so the outage is triggered by the power-deficit guard $P_{\mathrm{def}} > 22$. Because P_{def} is supplied in watts while the guard is written in kilowatts, this guard is true almost whenever there is any deficit, and the short return to $L = 2$ followed by a second outage shows chattering between the outage and critical-only states. Converting P_{def} to kilowatts before the chart would remove both effects (Section VI-G).

D. Test 4: Battery and Integrated-Model Response

Fig. 8 shows the battery scope from the integrated model over the first 0.22 s. After a start-up transient, the battery supplies about 75–160 A and 1.7–3.2 kW, but its terminal voltage settles at about 22–23 V instead of the 240 V nominal value, and its SoC starts at 51.6% instead of the 80% set in the script. Both differences point to the battery block parameters listed in Table III, and the trace is therefore treated as preliminary.

The integrated power scope (Fig. 9) covers about 0.21–0.31 s, which corresponds to 01:45–02:35 under the 0.12 s/h mapping. PV power is zero, as expected at night. The load power decreases smoothly from about 6.0 to 4.8 kW, and the deficit power reported by the EMS is equal to the load, which is correct when $P_{\mathrm{pv}} = 0$ in (15). Of the two HESS traces, one stays at 6.3–6.5 kW, slightly above the load, and the other rises slowly from 2.4 to 4.0 kW; the slow rise is consistent with the battery share following the 0.2 s low-pass filter in (16). The load in this window is below the 10.9 kW scheduled for that hour in Fig. 2 because the AC load sources are driven by a separate workspace profile rather than by the tier outputs of the load function. This window shows that storage supplies the night load, but it does not establish daytime PV transfer or the full-day energy balance.

E. Test 2: Inverter Waveforms

Fig. 10 shows the DC side of each H-bridge cell. The input voltage is a constant 200 V. The input current consists of PWM pulses whose envelope follows the rectified output current, reaching about 27 A, and it repeats every 10 ms, at twice the output frequency. Just after each zero crossing the current briefly reverses, reaching about −12 A, as reactive energy in the filter returns to the DC side. In the integrated system, the DC-link capacitors and the bus controller must therefore absorb a 100 Hz ripple current of this magnitude.

Fig. 11 compares the output voltage before and after the LC filter. Without the filter, the waveform shows the five levels 0, ±200 and ±400 V from (20) clearly, with PWM switching between adjacent levels; the closely spaced pulses correspond to the effective 4 kHz switching frequency of the phase-shifted carriers. The filtered voltage follows a sinusoidal envelope, but it carries large high-frequency oscillations with peaks of about ±505 V, 26% above the highest inverter level. A voltage above the 400 V DC sum can only arise from resonance, so the LC filter is underdamped and is excited by each switching edge. A damping resistor in series with the filter capacitor [27], or a lower filter cut-off frequency, is needed before the filter output can be used as the supply voltage.

The output current in Fig. 12 is nearly sinusoidal in both cases, with a peak of about 27 A and a small switching ripple of roughly ±0.5 A. The two traces are almost identical, which

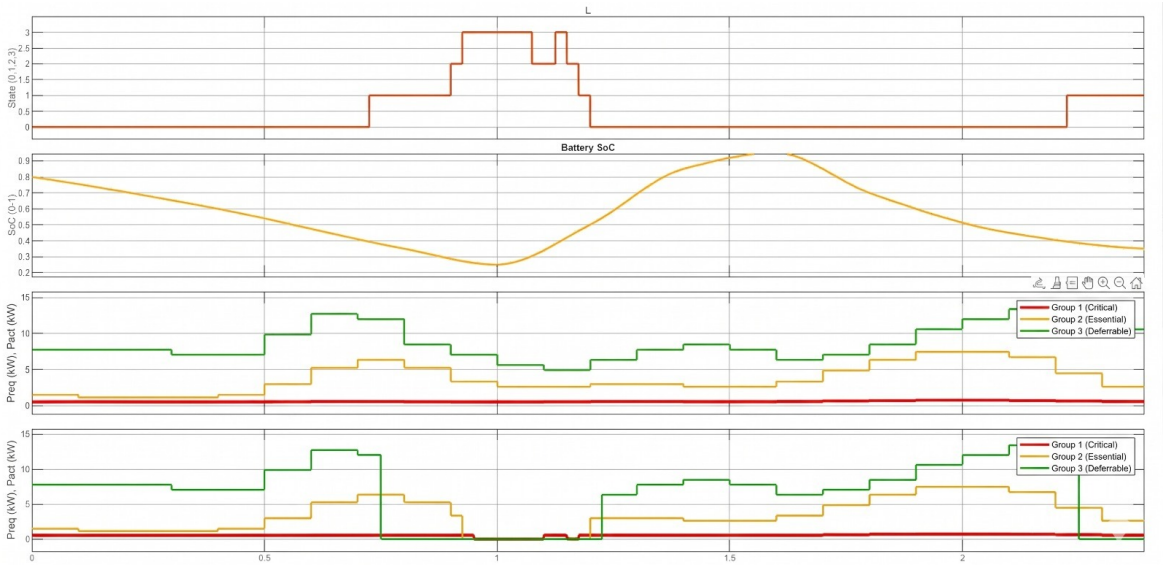


Fig. 7. Test 3: load-manager response to the benchmark SoC profile. From top: load-manager state L; battery SoC (0–1); requested tier power P_{req} (kW); active tier power P_{act} (kW). Time axis in simulated seconds.

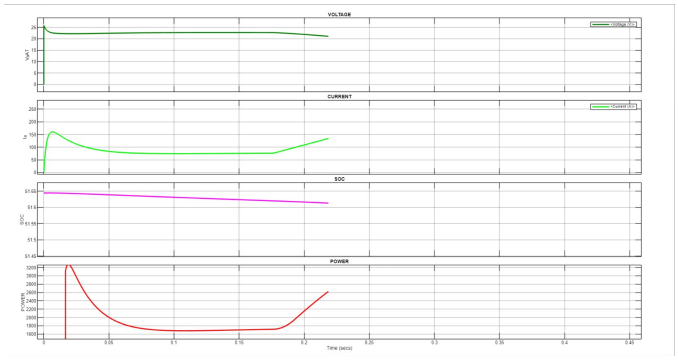


Fig. 8. Test 4: battery voltage (V), current (A), SoC (%) and power (W) in the integrated model. The voltage and initial SoC do not match the design values.

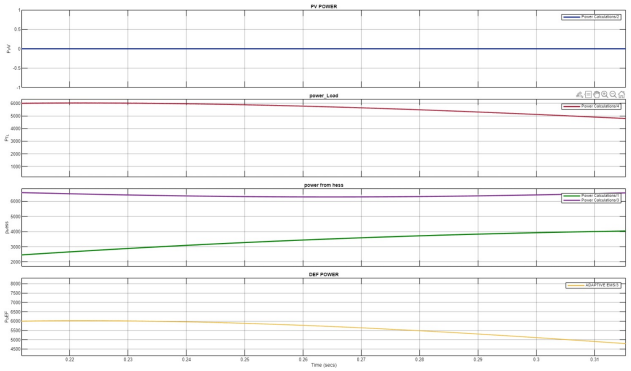


Fig. 9. Test 4: integrated power scope for 01:45–02:35 (simulated 0.21–0.31 s). From top: PV power, load power, HESS power and EMS deficit power, all in watts.

shows that the current is already smoothed by the series inductance and that the filter mainly affects the voltage.

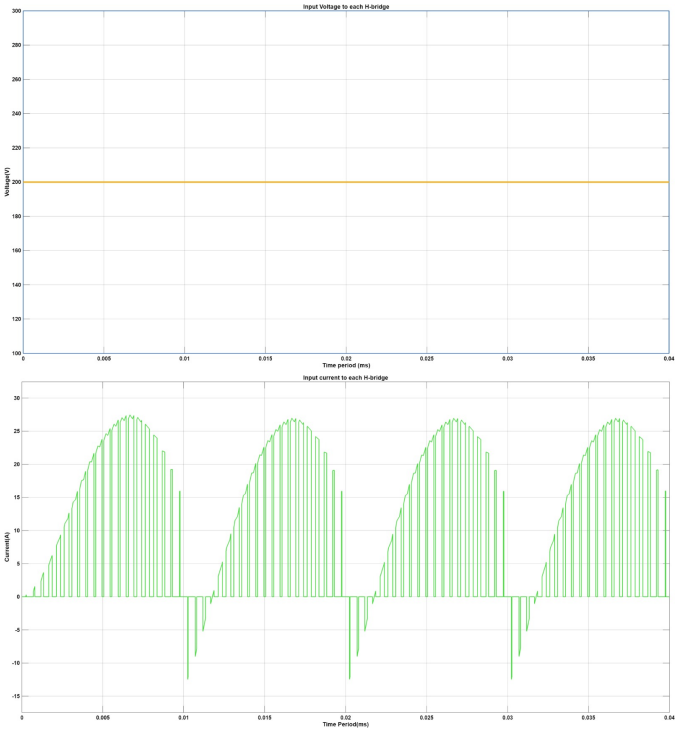
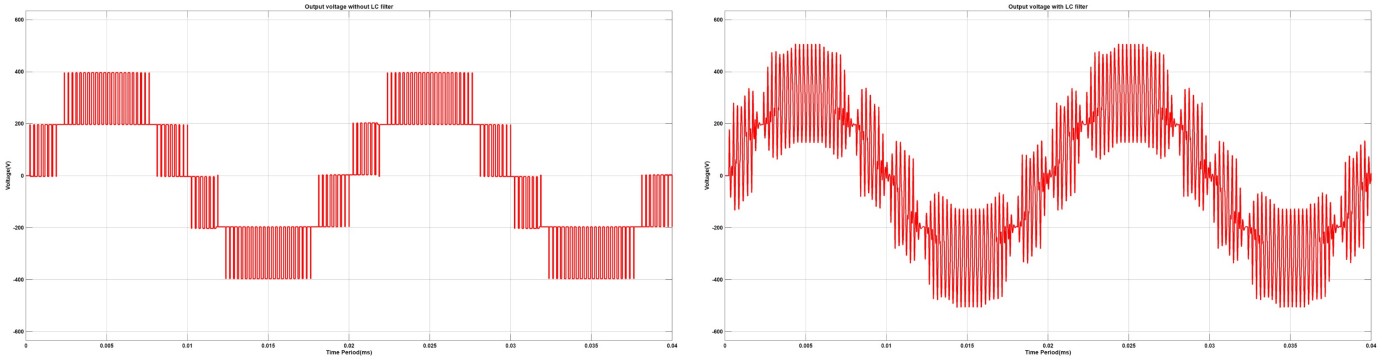


Fig. 10. Test 2: DC input voltage (top) and input current (bottom) of each H-bridge cell. Time axis in seconds.

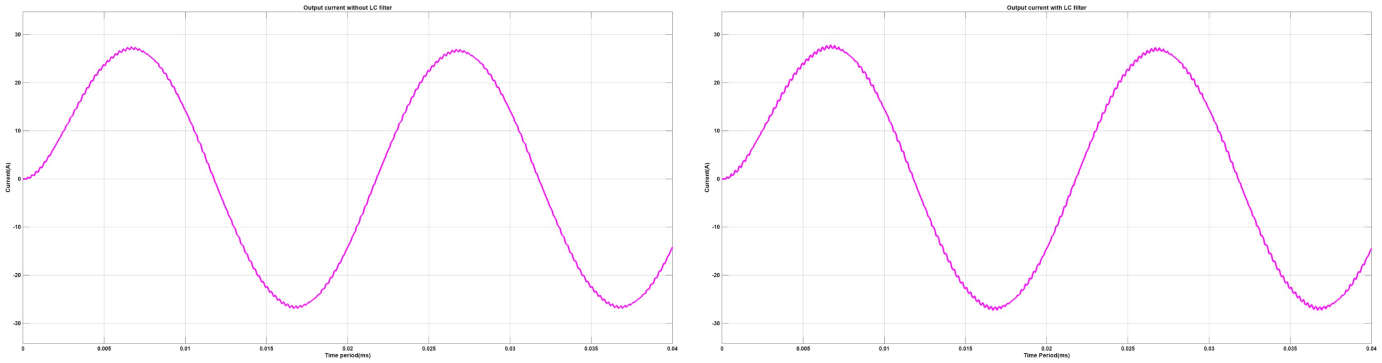
F. Test 2: Harmonic Analysis

Fig. 13 shows the FFT spectra and the harmonic magnitudes up to 1 kHz (the 20th harmonic). The voltage fundamental is about 318 V peak (225 V RMS) and the current fundamental about 26.9 A peak (19 A RMS), so the test load draws about 4.3 kW at an equivalent resistance of about 11.8 Ω. The largest



(a) Without LC filter (b) With LC filter

Fig. 11. Test 2: CHB inverter output voltage (V) over two 50 Hz cycles. (a) Five-level PWM waveform with levels 0, ±200 and ±400 V. (b) Filtered voltage showing resonant overshoot to about ±505 V. Time axis in seconds.



(a) Without LC filter (b) With LC filter

Fig. 12. Test 2: CHB inverter output current (A) over two 50 Hz cycles, (a) without and (b) with the LC filter. Time axis in seconds.

voltage harmonics are the 3rd, 7th, 9th and 11th, each below about 1.5 V RMS, or under 0.7% of the fundamental. Using (24) over this range gives a voltage THD of 0.93% and a current THD of 0.26%.

These values are well within the IEEE Std 519 limits for systems at or below 1 kV, which are 5% for individual harmonics and 8% for voltage THD [26]. They describe the low-order harmonic content only. The standard counts harmonics up to the 50th order (2.5 kHz), and the switching components visible in Fig. 11(b) lie above the 1 kHz analysis window. The THD should therefore be recomputed up to at least the 50th harmonic, after the filter is damped, before compliance is claimed.

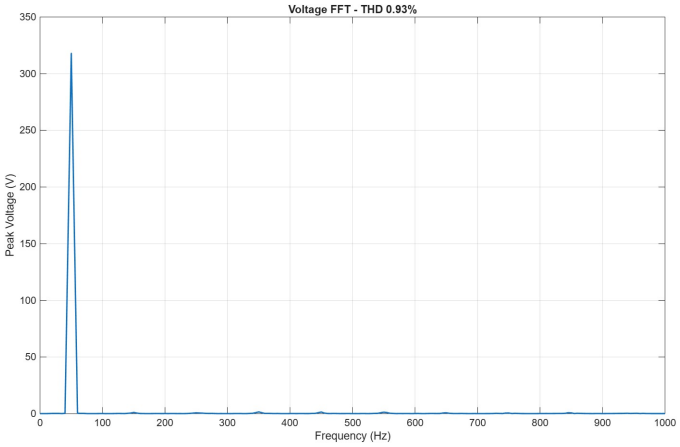
G. Model Consistency and Limitations

Comparing the supplied model with the initialisation script and the results above identifies the following items, which explain the preliminary nature of Tests 3 and 4 and must be resolved before a full-day closed-loop run:

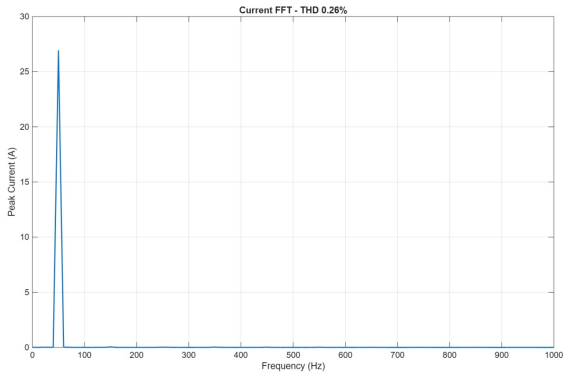
- The battery and supercapacitor block capacities (Table III) do not equal the time-scaled design values from (12); the battery block corresponds to 50 Ah instead of 541.7 Ah, consistent with the battery voltage and SoC seen in Fig. 8.
- The EMS filter uses $\tau = 0.2$ s, which under the 0.12 s/h mapping equals about 100 real minutes. The script recommends 2 ms (one real minute) for the compressed day, so that the battery, not the supercapacitor, carries the sustained energy.
- The supervisory SoC input is the benchmark profile, so the EMS and load manager are not yet driven by the battery's own SoC. The adaptive function (17) expects

TABLE IX
PERFORMANCE INDICES FOR THE TESTED CASES

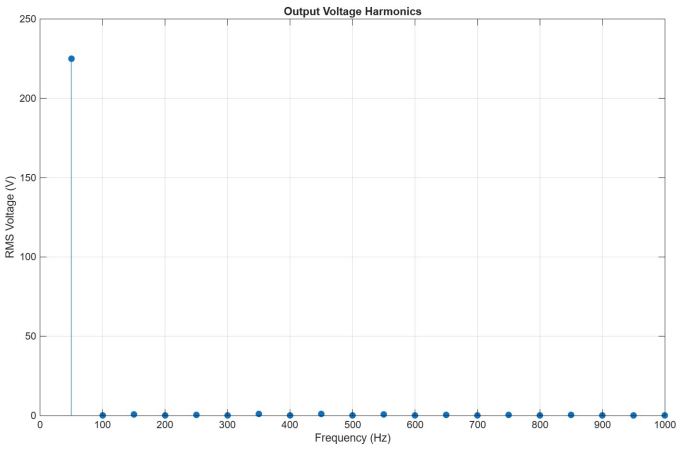
Index	Value	Source
Peak irradiance / peak cell temperature	950 W/m^2 / 62 °C	Test 1
Peak PV power per unit (simulated / (25))	15.4 / 15.5 kW	Test 1
PV voltage / duty ratio near noon	200 V / 0.50	Test 1
Daily PV energy, one / two units	101.6 / 203 kWh	Test 1
Daily demand, total / firm	211.6 / 117.8 kWh	Load model
Peak demand (21:00–23:00)	16.5 kW	Load model
Energy coverage ratio ρ	0.96	Test 1
Peak DC-bus voltage / deviation	405.5 V / 1.4%	Test 1
Shed / restore thresholds observed	0.40, 0.30 / $+d$	Test 3
Battery terminal voltage in integrated model	22–23 V	Test 4
Inverter output, fundamental	225 V, 19 A RMS	Test 2
Filtered voltage peak (overshoot)	±505 V	Test 2
Voltage / current THD (to 20th harmonic)	0.93% / 0.26%	Test 2



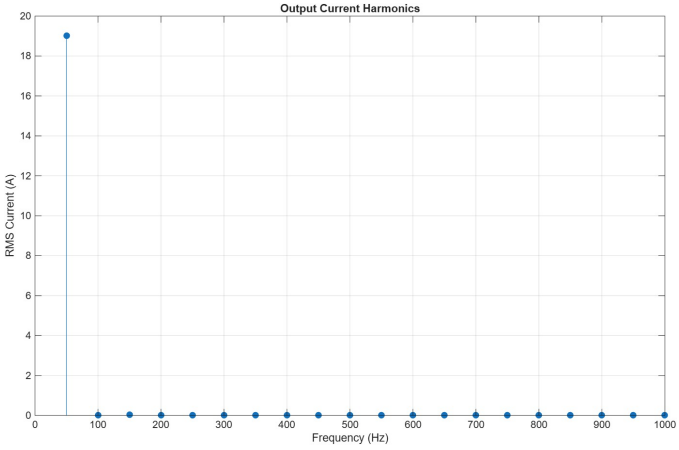
(a) Voltage FFT, THD 0.93%



(b) Current FFT, THD 0.26%



(c) Voltage harmonics (RMS)



(d) Current harmonics (RMS)

Fig. 13. Test 2: harmonic analysis of the inverter output up to 1 kHz. (a), (b) FFT spectra (peak values); (c), (d) RMS magnitude of each harmonic of the 50 Hz fundamental.

SoC in percent while the load manager expects a fraction, and P_{def} is in watts while the outage guards (22 and 10) are written in kilowatts, which causes the outages seen in Fig. 7.

- The AC load sources use a workspace profile instead of the tier powers of the load function, so the load seen by the HESS does not match the schedule in Fig. 2.
- The LC filter is underdamped (Fig. 11(b)), and the THD has been evaluated only up to 1 kHz.
- The DC-link capacitance (300 µF) is much smaller than the 25 mF design value from (14), the PV boost capacitors are 100 µF rather than 270 µF, and the inverter efficiency of 0.96 is not yet applied in (15).

VII. Conclusion

This paper presented a MATLAB/Simulink design of a solar-powered microgrid for 12 flats, combining a 38.4 kW PV array with P&O MPPT, a 400 V DC bus, battery–supercapacitor storage with SoC-aware low-pass-filter sharing, a priority load manager and a five-level CHB inverter. Two PV units supply about 203 kWh/day, 96% of the 211.6 kWh/day demand, and the simulated peak of 15.4 kW per unit agrees with the temperature-corrected estimate to within 1%. The DC bus stays within 1.4% of 400 V. The inverter gives voltage and current THD of 0.93% and 0.26% up to the 20th harmonic, and the load manager sheds and restores tiers in priority order with hysteresis.

The results also show that generation and demand are poorly aligned in time: the evening peak of 16.5 kW occurs after sunset, so about 68 kWh of firm night demand must come from storage. The 130 kWh battery, with 91 kWh usable in its SoC window, covers this requirement, while the supercapacitor takes the fast transients.

Future work will correct the parameter and unit mismatches listed in Section VI-G, damp the LC filter, extend the harmonic analysis to the 50th order, connect the battery's own SoC to the EMS, and run the integrated model over a full clear and partly cloudy day before hardware validation; grid-connected operation would also need to meet IEEE Std 1547 [28].

Acknowledgment

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